

A CONCEPTUAL INTRODUCTION TO HAMILTONIAN MONTE CARLO

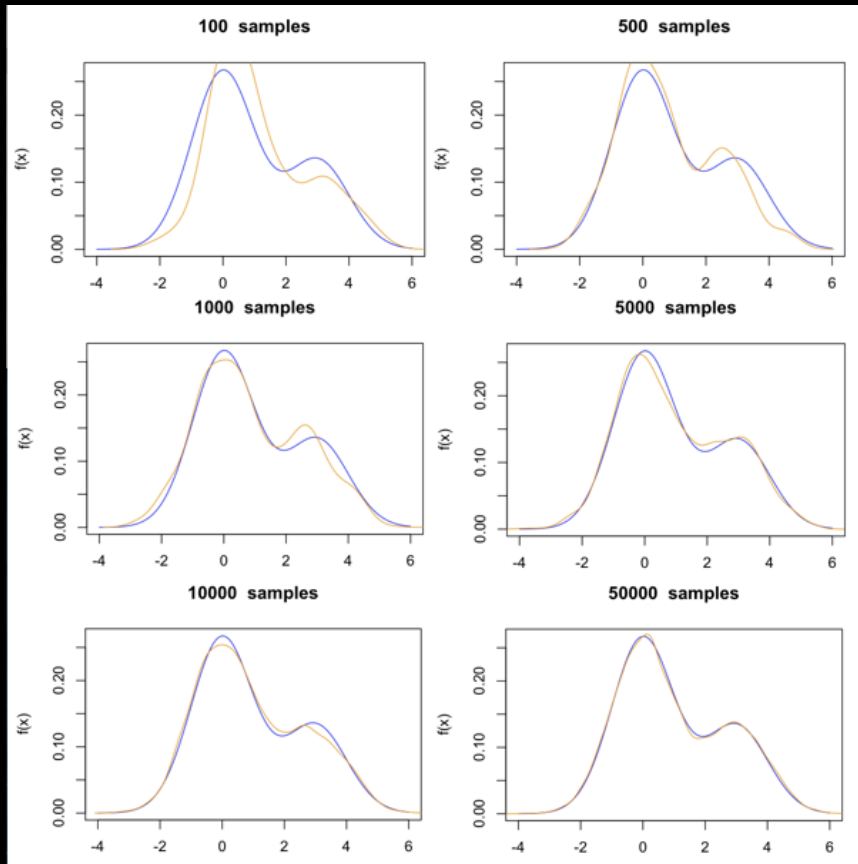
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Sampling From Probability Distributions with MCMC-s

- Markov chain Monte Carlo (MCMC) algorithms are used to iteratively draw samples from PDF-s.
- High-dimensional PDF-s are difficult to handle analytically.
- MCMC is used in astronomy, medical sciences, finance etc.
- Metropolis-Hastings MCMC (1953 for stat. phys., 1970 for general) - first and the most popular variant of MCMC.



(Markov chain Monte Carlo, 2025)
An example of probability distribution estimation using MCMC.

Historical Overview of HMC

- Invented for 1980s Lattice Quantum Chromodynamics (L-QCD) (Duane et al., 1987)
 - a study of quarks: *a non-perturbative gauge theory*
 - “Bayesian Learning for Neural Networks” PhD thesis (Neal, 1995)
 - “The Geometric Foundations of Hamiltonian Monte Carlo” (Betancourt et al., 2014)
 - differential geometry
 - developed a rigorous mathematical understanding (earlier analysis was based on fragile heuristics)
 - This article (Betancourt, 2017)
 - a simplified overview of the rigorous mathematical justification of the HMC
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Estimating Expectations

Suppose we want to estimate the expected values of a probability density function $\pi(q)$, $q \in Q$ using MCMC. Analytical expectation for a function defined on the same sample space is

$$\mathbb{E}_{\pi}[f] = \int_Q \pi(q) f(q) dq.$$

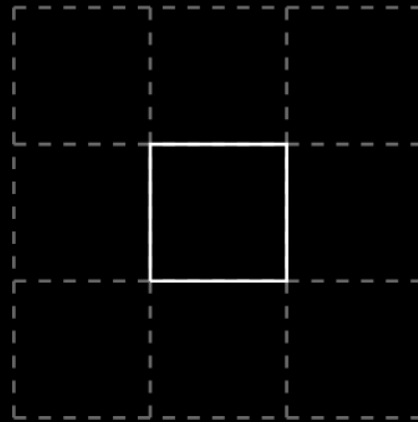
To sample the pdf most accurately, the MCMC iteration must imitate the behaviour of the term $\pi(q) dq$ across the sample space.

The term $\pi(q) dq$, the probability mass, gives the most important contribution!

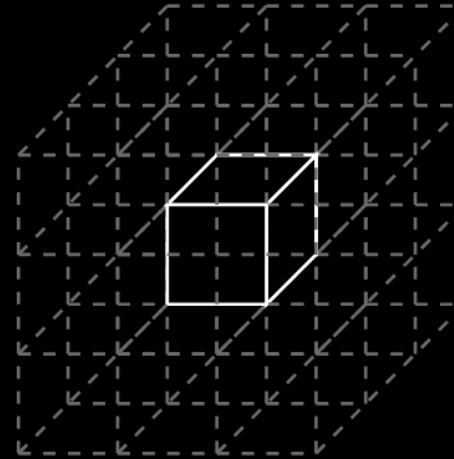
The Geometry of High-Dimensional Spaces (1/3)



(a)



(b)

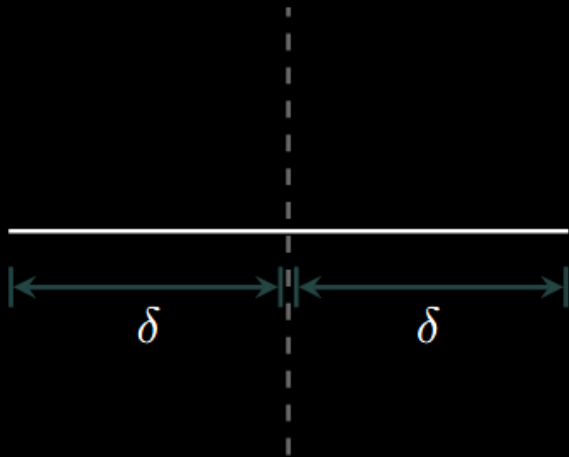


(c)

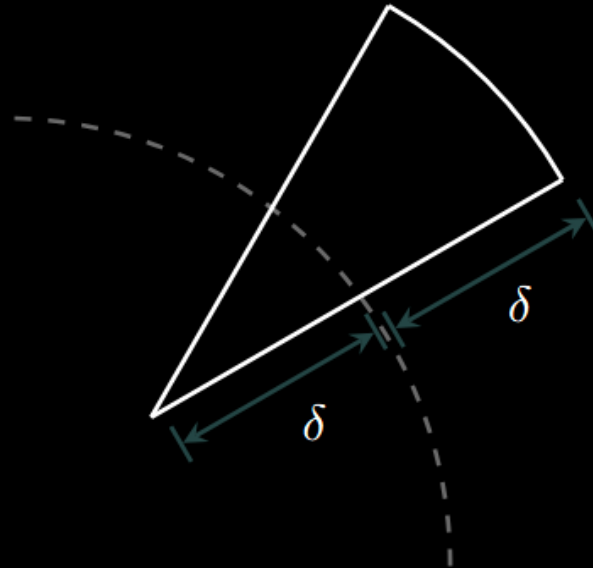
(Betancourt, 2017)

The number of neighbors and the size of the infinitesimal shell from an infinitesimal center unit cell grows exponentially with the dimension.

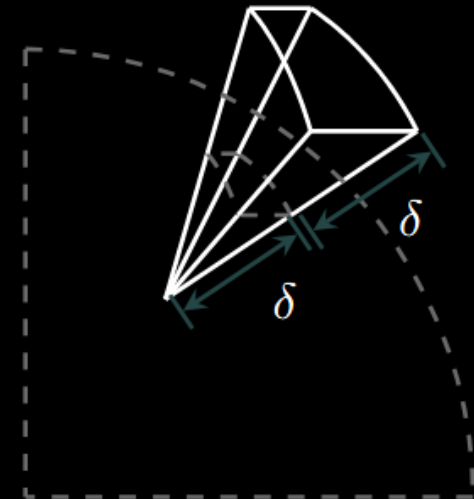
The Geometry of High-Dimensional Spaces (2/3)



(a)



(b)



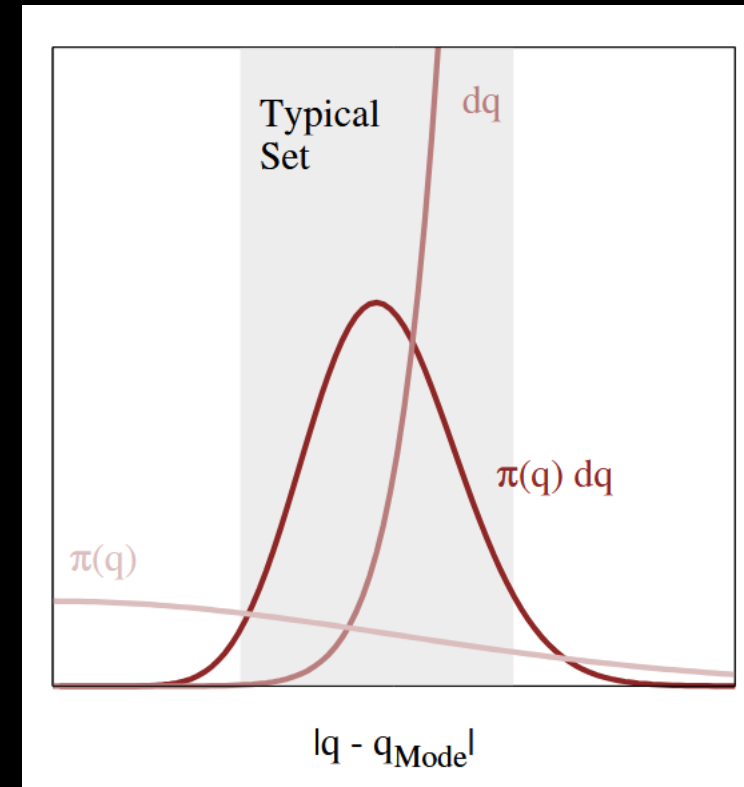
(c)

(Betancourt, 2017)

In general spherical coordinates, the size of each individual infinitesimal unit also increases with the distance from the origin.

The Geometry of High-Dimensional Spaces (3/3)

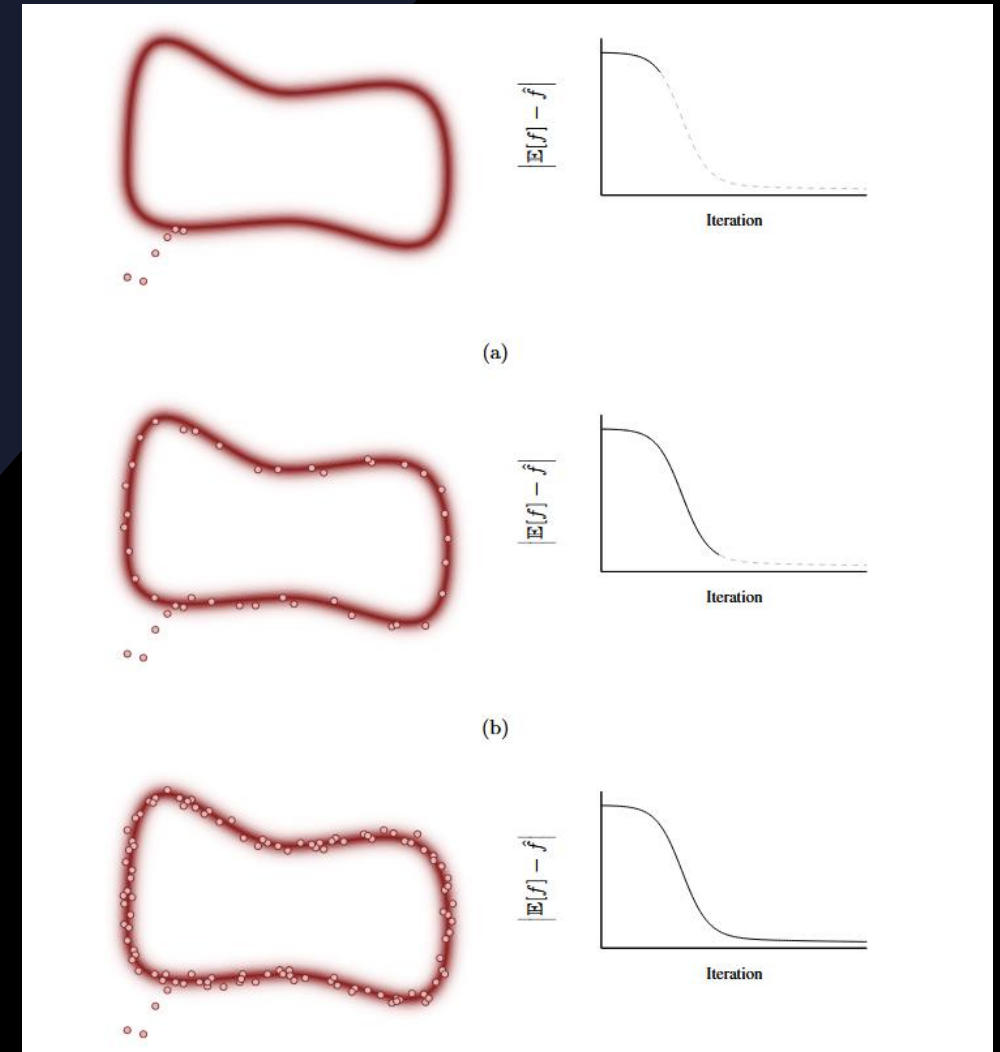
- The typical set (term coined by Betancourt) is the set of points inside the sample space where the probability mass term $\pi(q)dq$ contribution to the expectation is the highest.
- Typical sets tend to form closed contours around the modes.
- Naive finite discretization methods fail to be optimal ($\mathcal{O}(N^D)$).
- Optimal and accurate iterators have to orbit the typical set.



(Betancourt, 2017)
 dq is the infinitesimal shell volume.

Typical Sets of Probability Density Functions

- Typical sets are around the modes of PDF-s.
- MCMC will produce a chain of samples according to some Markov chain transition rule.
- Ideal MCMC must orbit the typical set.



(Betancourt, 2017)

Markov Chain and Markov Property

- Markov chain is a sequence of random variables $X_1, X_2, X_3, \dots$ where the probability of moving to a new state depends only on the previous state (the previous realization).
- The ergodic theorem implies that some Markov chains tend to act asymptotically and offers a rigorous proof that MCMC iteration parameters converge to PDF-s true parameters.

$$\mathbf{P}\{X_{n+1} = x | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\} = \mathbf{P}\{X_{n+1} = x | X_n = x_n\}.$$

$$P = \begin{array}{c} \text{today} \\ \text{sunny} \\ \text{rainy} \end{array} \overbrace{\begin{array}{cc} \text{tomorrow} & \\ \text{sunny} & \text{rainy} \\ \begin{pmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{pmatrix} \end{array}}$$

$$\mathbf{P}\{\text{"tomorrow sunny"} | \text{"today sunny"}\} = (P)_{11} = 0.3$$

$$\mathbf{P}\{\text{"tomorrow rainy"} | \text{"today sunny"}\} = (P)_{12} = 0.7$$

$$\lim_{n \rightarrow \infty} P^n = \begin{array}{c} \text{today} \\ \text{sunny} \\ \text{rainy} \end{array} \overbrace{\begin{array}{cc} \text{in forever} & \\ \text{sunny} & \text{rainy} \\ \begin{pmatrix} 0.125 & 0.875 \\ 0.125 & 0.875 \end{pmatrix} \end{array}}$$

$$\mathbf{P}\{\text{"sunny on average"}\} = 0.125$$

$$\mathbf{P}\{\text{"rainy on average"}\} = 0.875$$

General MCMC Concept

- Iteratively, a Markov chain of N samples over the sample space is constructed with an initial guess q_0 and the conditional Markov transition rule:

$$q_{n+1} \sim \mathbb{T}(q_{n+1}|q_n)$$

- A Markov chain samples $\{q_0, q_1, \dots, q_N\}$ are used to estimate $E_\pi[f]$:

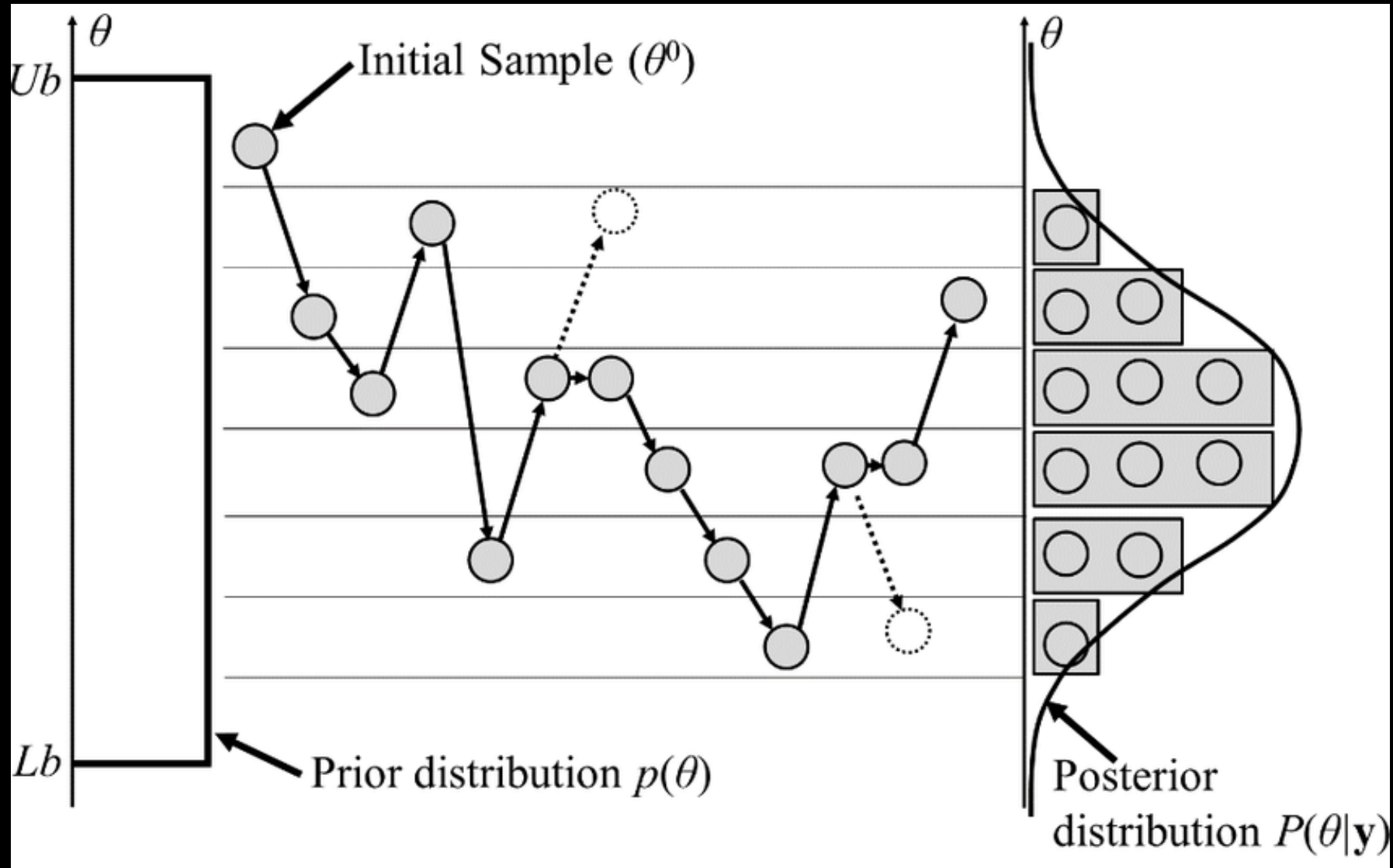
$$\hat{f}_N = \frac{1}{N} \sum_{n=0}^N f(q_n)$$

- According to the ergodic theorem of MCMC, this will converge to the true value:

$$\lim_{n \rightarrow \infty} \hat{f} = \mathbb{E}_\pi[f]$$

Random-Walk Metropolis-Hastings

- In Random-Walk Metropolis-Hastings the new proposal is generated by a Gaussian kernel function.
 - The acceptance probability depends on the ratio of the PDF values evaluated on the proposal and previous sample, respectively.
 - The new sample will be accepted or rejected depending on that probability.
 - This transition rule doesn't represent the typical set as the dimension grows.
- $q_0 \in Q$ (initialize a random point in sample space)
for $0 \leq n \leq N - 1$ do
 $q^* = q_n + \Delta, \quad \Delta \sim \mathcal{N}(0, \sigma^2)$
 $\alpha = \min \left\{ 1, \frac{\pi(q^*)}{\pi(q_n)} \right\}$
 $\beta \sim \text{Unif}(0, 1)$
 $q_{n+1} = \begin{cases} q^*, & \text{if } \beta \leq \alpha \\ q_n, & \text{else} \end{cases}.$
-



(Metropolis-Hastings algorithm, 2025)

Illustrative example of Metropolis-Hastings MCMC used for Bayesian inference.

Hamiltonian Monte Carlo Algorithm (1/4)

- HMC treats $\pi(q)$ as if it is a PDF of a conservative potential field:
 - potential energy $U(q) = -\log \pi(q)$
 - kinetic energy $K(p) = \frac{1}{2}p^\top M^{-1}p + \text{const}$, $p \sim \mathcal{N}(0, M^{-1})$
 - Hamiltonian $H(q, p) = K(p) + U(q)$
 - Hamiltonian dynamics is simulated for each sample proposal before the accept-reject test.
 - While Metropolis-Hastings “diffuses” slowly in the sample space, the HMC goes to the phase space of generalized coordinates and the auxiliary momenta, where it simulates trajectories that might move fast.
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Hamiltonian Monte Carlo Algorithm (2/4)

- Hamilton's equations are being simulated in phase space of sample coordinates and momenta.
- Symplectic (energy conserving) leapfrog integrator is being used for simulating dynamics.
- The new sample will be added depending on the acceptance probability.

$$\dot{p} = -\frac{\partial H}{\partial q}$$

$$\dot{q} = \frac{\partial H}{\partial p}$$

$$p_{i+\frac{1}{2}} = p_i + \frac{\Delta t}{2} \nabla U(q_i)$$

$$q_{i+1} = q_i + M^{-1} p_{i+\frac{1}{2}}$$

$$p_{i+1} = p_{i+\frac{1}{2}} + \frac{\Delta t}{2} \nabla U(q_{i+1})$$

$q_0 \in Q$, (initialize a random point in sample space)

for $0 \leq n \leq K - 1$ do

$$q_0^* = q_n, \quad 0 \leq i \leq L$$

$$p_0 \sim \mathcal{N}(0, M^{-1})$$

for $0 \leq i \leq L - 1$ do

$$p_{i+\frac{1}{2}} = p_i + \frac{\Delta t}{2} \nabla U(q_i)$$

$$q_{i+1}^* = q_i^* + M^{-1} p_{i+\frac{1}{2}}$$

$$p_{i+1} = p_{i+\frac{1}{2}} + \frac{\Delta t}{2} \nabla U(q_{i+1})$$

$$\alpha = \min\{1, \exp(-H(q_L^*, -p_L) + H(q_n, p_0))\}$$

$$\beta \sim \text{Unif}(0, 1)$$

$$q_{n+1} = \begin{cases} q_L^*, & \text{if } \beta \leq \alpha, \\ q_n, & \text{else.} \end{cases}$$

Hamiltonian Monte Carlo Algorithm (3/4)

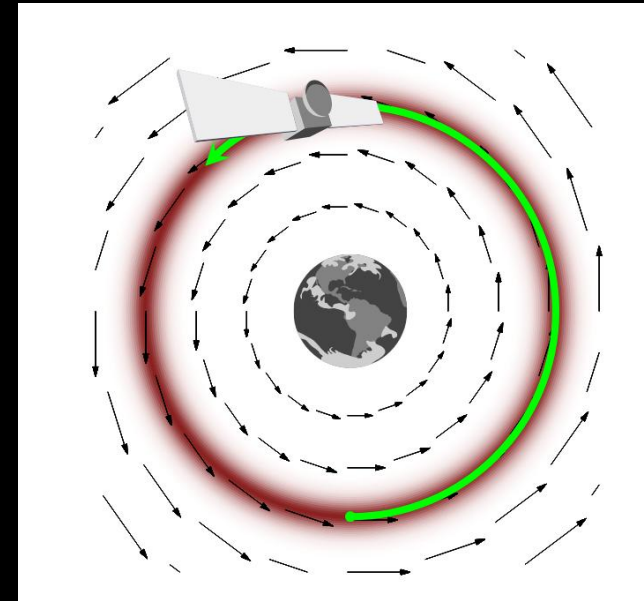
- After each integration loop, an acceptance probability is calculated depending on the value of the current Hamiltonian will be calculated.
- The new sample will be accepted or rejected depending on that probability.

Hamiltonian Monte Carlo Algorithm (4/4)

Input: $K, L, \theta^{(0)}, \epsilon, \mathcal{L}$
for $k = 1, \dots, K$ **do**
 Sample $r_0 \sim \mathcal{N}(0, I)$.
 Set $\theta_0 \leftarrow \theta^{(k-1)}$.
 for $i = 1, \dots, L$ **do**
 Set $r_i \leftarrow r_{i-1} + \frac{\epsilon}{2} \nabla_{\theta} U(\theta) |_{\theta=\theta_{i-1}}$.
 Set $\theta_i \leftarrow \theta_{i-1} + \epsilon r_i$.
 Set $r_i \leftarrow r_i + \frac{\epsilon}{2} \nabla_{\theta} U(\theta) |_{\theta=\theta_i}$.
 With probability
 $\min\{1, \exp(-Ha(\theta_L, -r_L) + Ha(\theta_0, r_0))\}$,
 Set $(\theta^{(k)}, r^{(k)}) = (\theta_L, -r_L)$.

(Amrii et al., 2025)

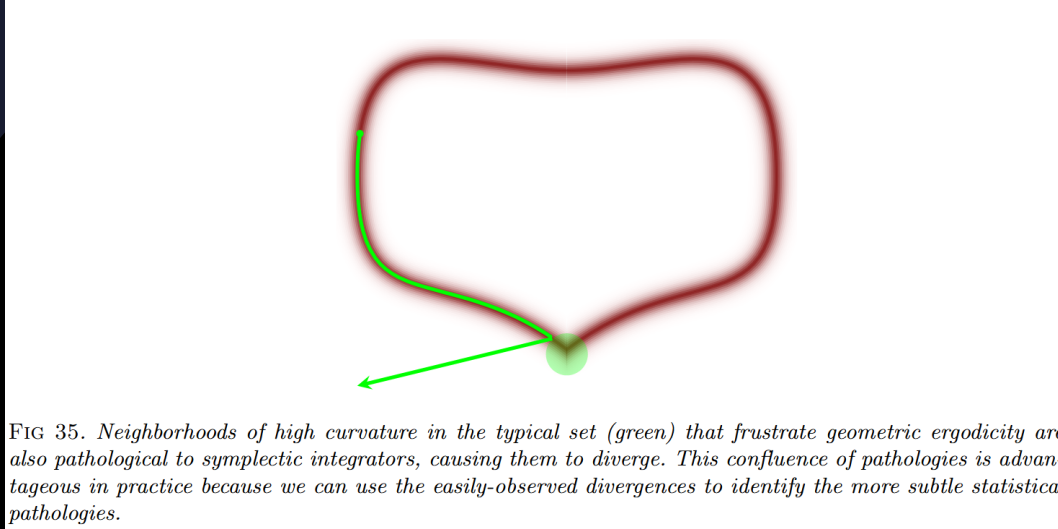
Example of a HMC loop definition in an article.



(Betancourt, 2017)

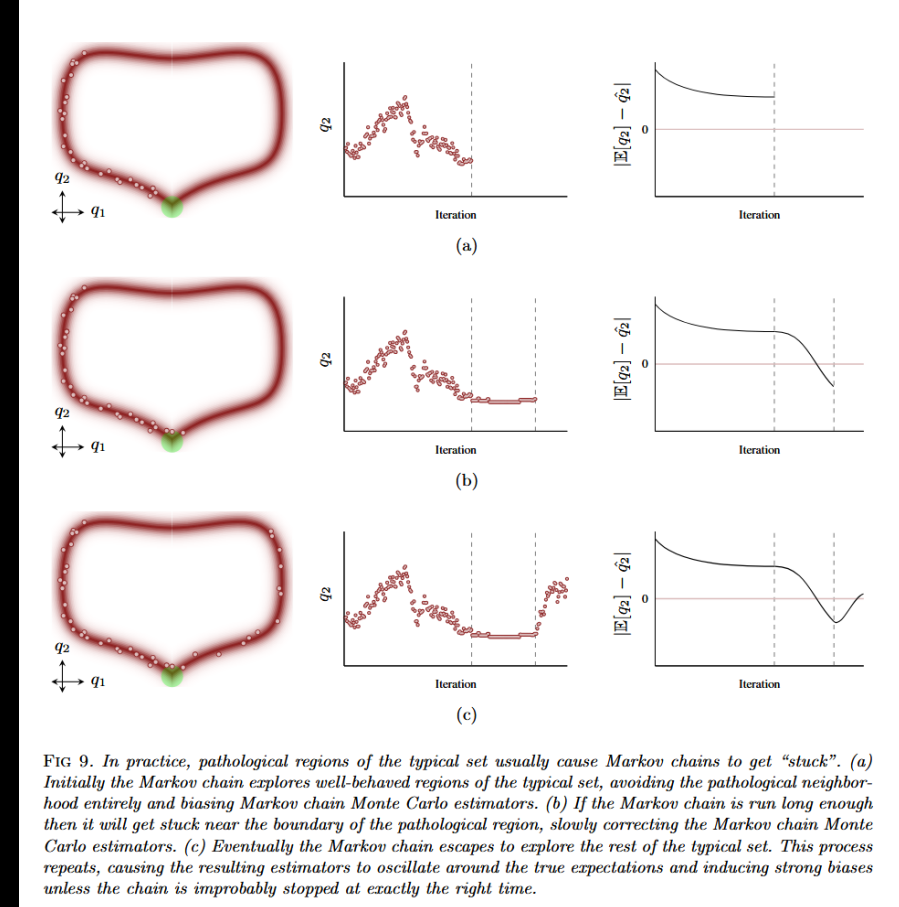
Generating a new HMC sample alongside the typical set is analogous with orbital motion.

Problematic Subsets and Pathological Behaviour



(Betancourt, 2017)

Divergences in the samples of HMC help to diagnose places of high curvature...

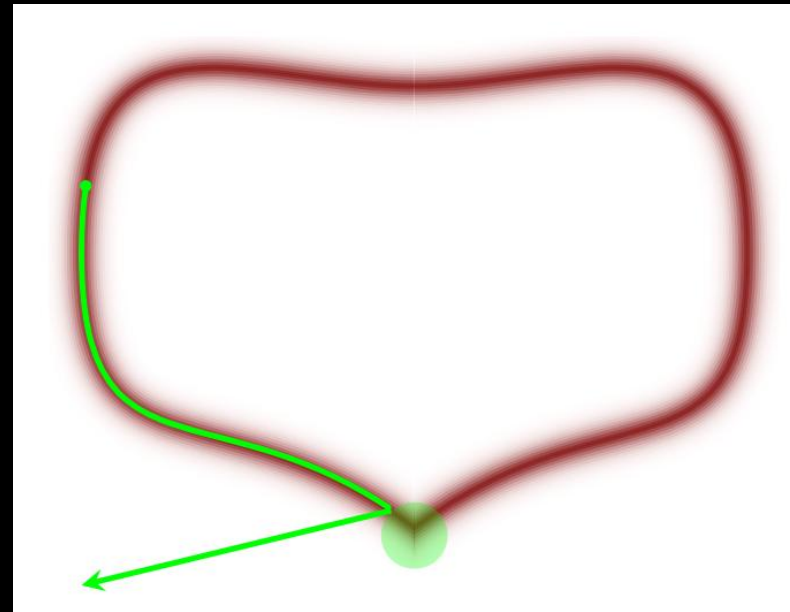


(Betancourt, 2017)

...whereas non-dynamical MCMC sampler will get stuck.

Insights of HMC Behaviour on Typical Set

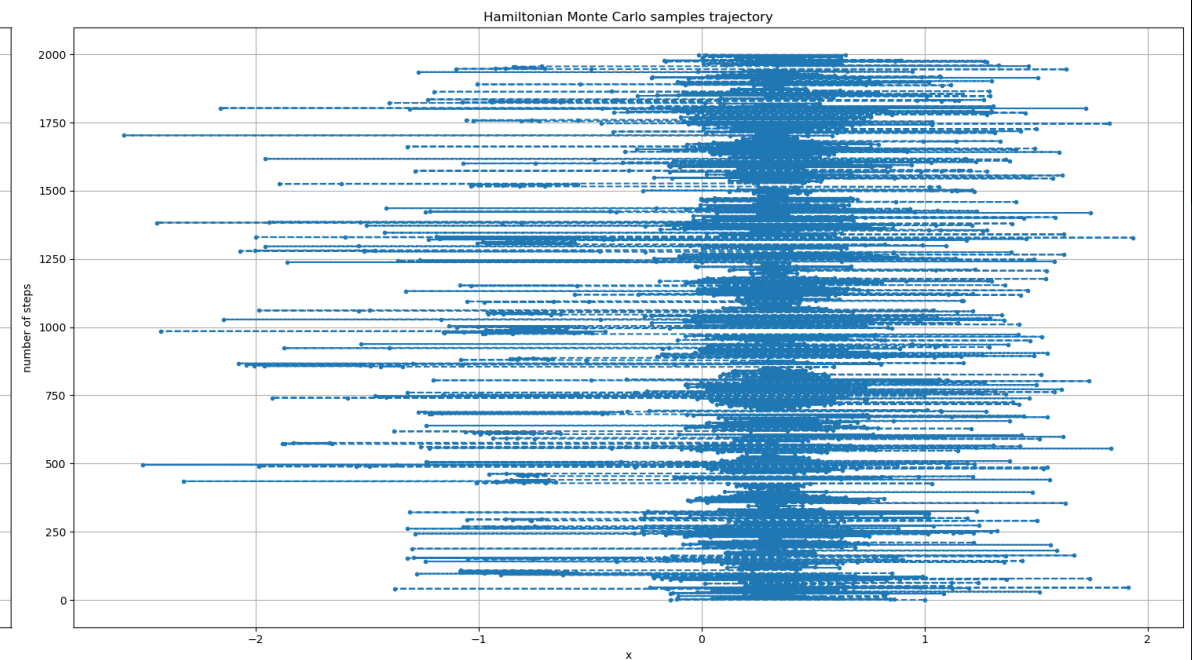
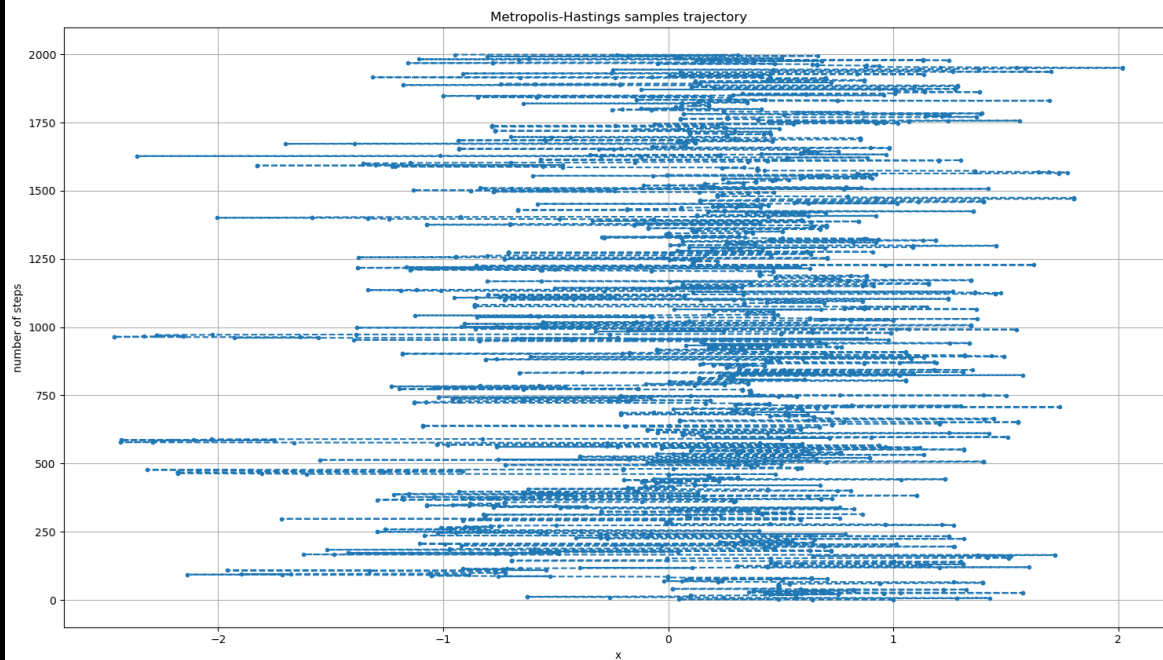
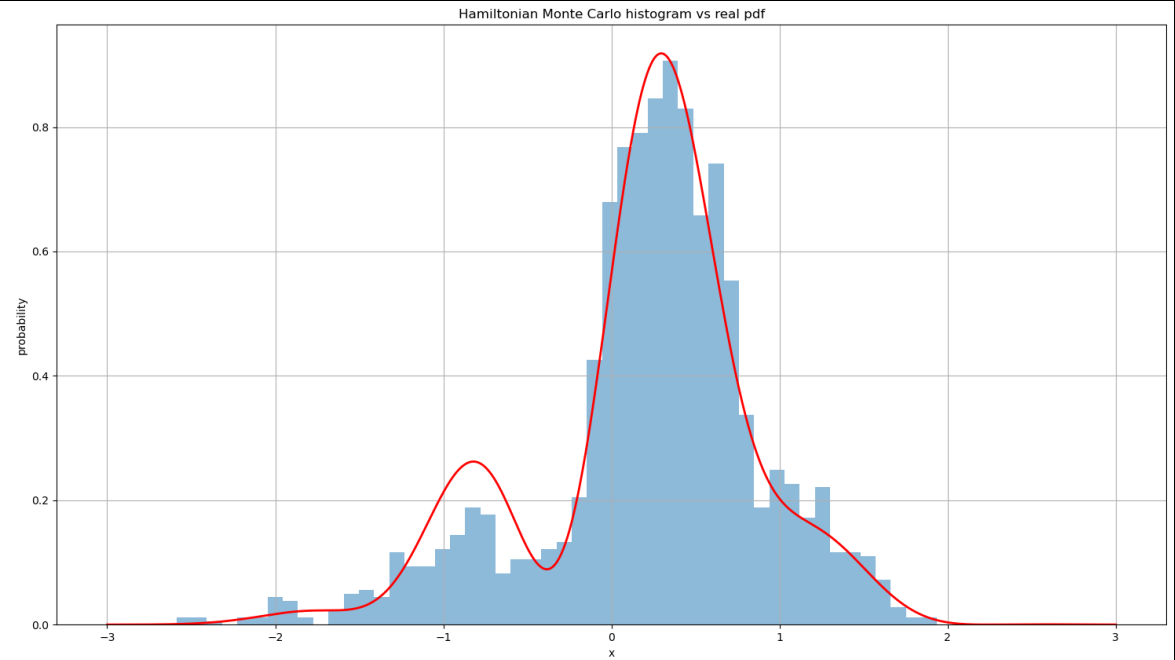
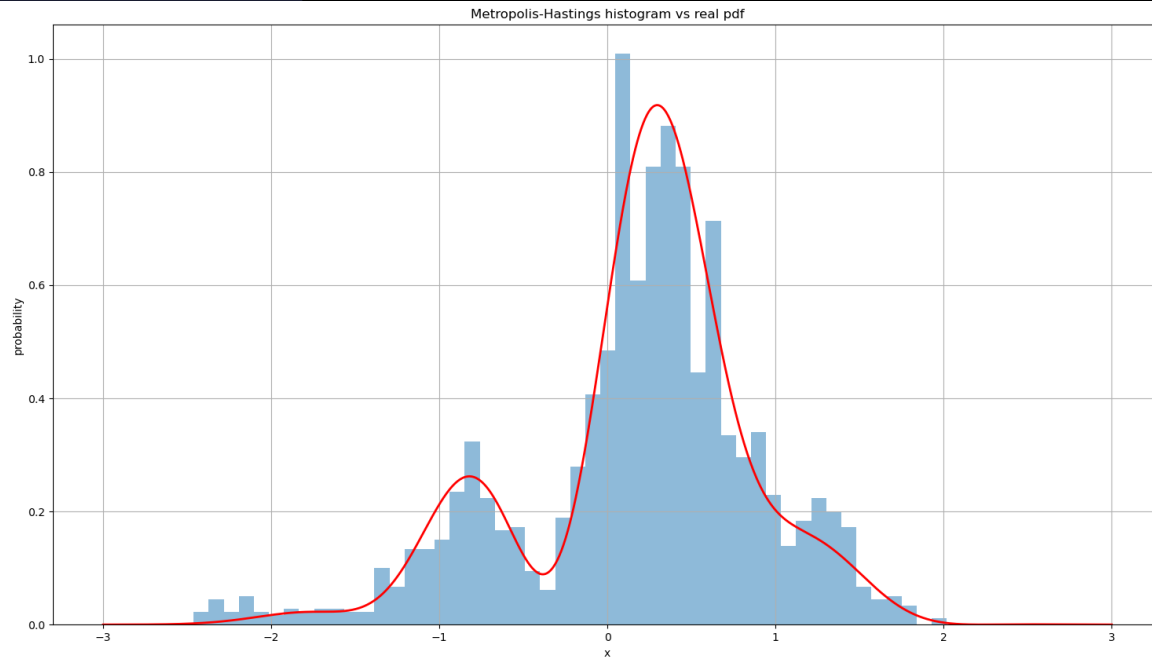
- HMC orbits the typical set accurately and doesn't over-represent the mode.
- Phase space volume preserving tendency of Hamiltonian dynamics is a useful property to generate accurate samples.
- Divergences in the sample space of HMC help to diagnose places of high curvature.



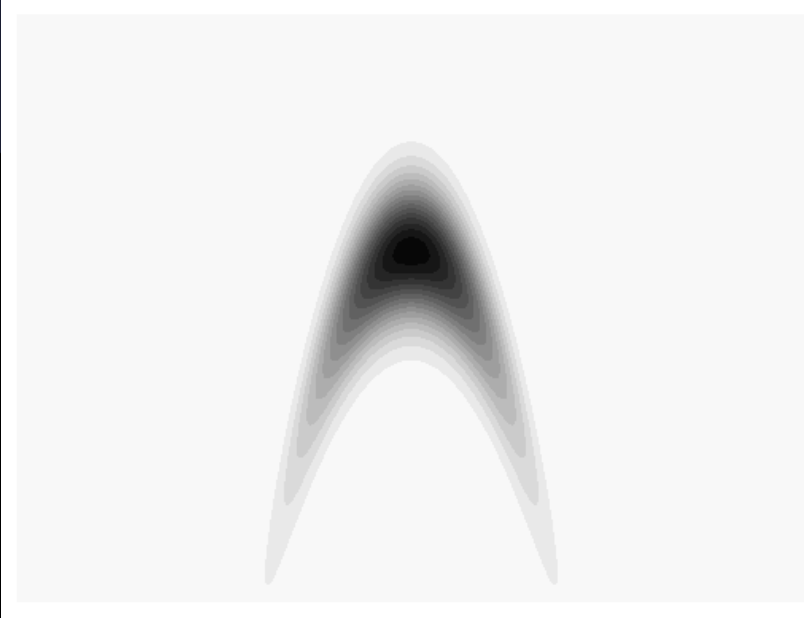
(Betancourt, 2017)

HMC samplers “shoot off” from the typical sets with problematic geometry.

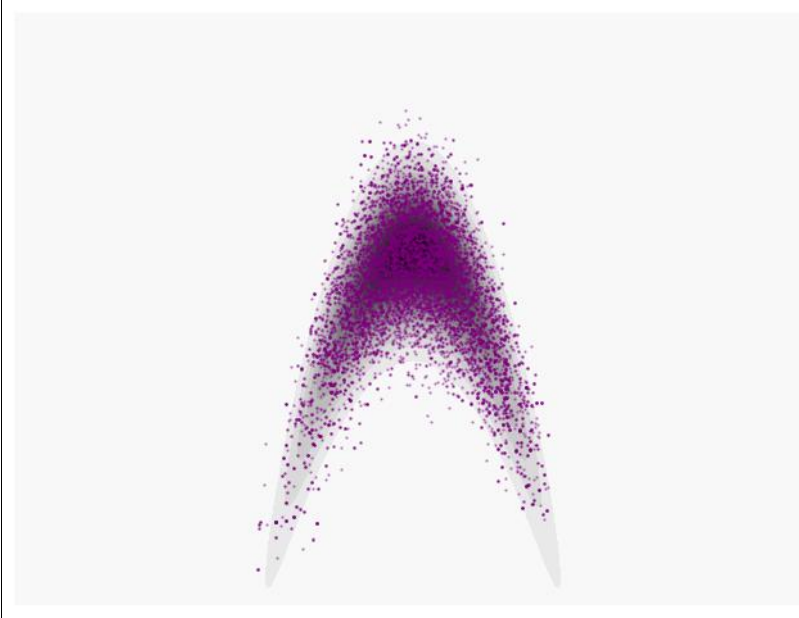
HMC vs Metropolis-Hastings Example: 1D PDF (made with python, 2000 samples)



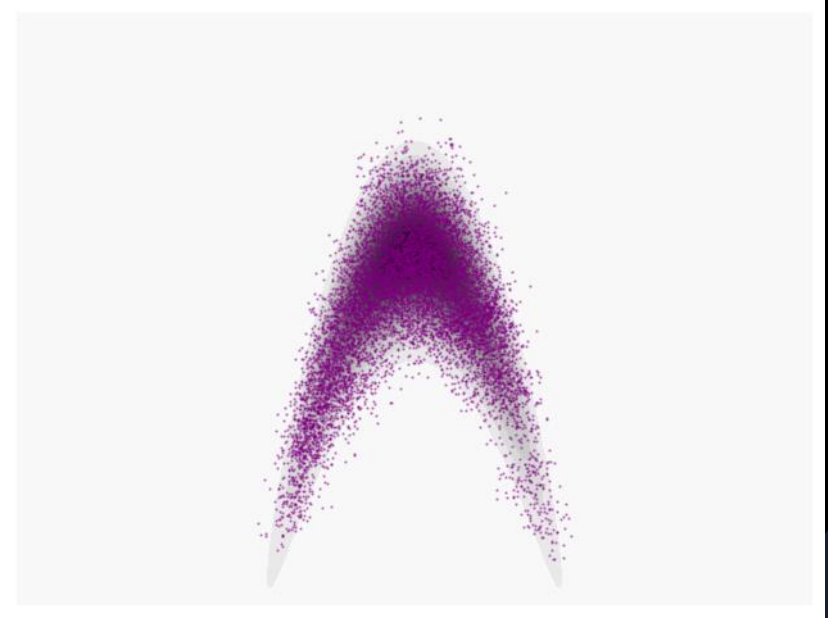
HMC vs Metropolis-Hastings Example: a 2D banana-shaped PDF (made with python, 2000 samples)



True PDF contour-plot.



Metropolis-Hastings samples.



HMC samples.

Examples of uses of HMC

- Binary neutron star parameter estimation [arXiv:2505.02589]
 - accelerated by a deep neural network
 - 12-D parameter sample-space
 - Parameter estimation for high-dimensional PDEs [arXiv:2509.14054]
 - parameter estimation for a 50-dimensional heat equation
 - Bayesian Neural Networks [arXiv:2006.12024]
 - *HMC can be employed to incorporate gradient information into the iterative behaviour.*
 - Computational biology
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References

- Betancourt, M. (2017). A Conceptual Introduction to Hamiltonian Monte Carlo. (arXiv:1701.02434)
 - Markov chain Monte Carlo. (2025). Wikipedia.
 - Metropolis–Hastings algorithm. (2025). Wikipedia.
 - Amri, A., Elvira, V., & Wilson, A. L. (2025). Particle Hamiltonian Monte Carlo. (arXiv:2504.09875)
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Thank you for listening!
